

Jack Smith

Computer Science Student · AI, Graphics & Interactive Systems

CONTACT

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TECHNICAL SKILLS

PROGRAMMING

Python, TypeScript, JavaScript, C#,
Haskell, Ruby, SQL

AI & ML

PyTorch, CNNs, transfer learning,
reinforcement learning, generative AI,
computer vision, on-device inference

GRAPHICS & 3D

Unity, Three.js, shaders, rasterization,
Blender

WEB & FULL-STACK

React, Next.js, SvelteKit, Node.js,
Flask, Express, Rails, Tailwind, GSAP

DATA & DEVOPS

PostgreSQL, Firebase, Cloudflare
Workers, Docker, Vercel, Git, CLI and
MCP tooling

LANGUAGES

English Native
Arabic Intermediate

PROFILE

Computer Science student at the University of Sheffield, graduating in 2026, on track for first-class honours, with a strong builder's mindset and a focus on AI, computer graphics, and interactive systems. I design and ship technically demanding software, from full-stack apps for real clients to low-level graphics pipelines and generative AI tools, and I am drawn to problems where AI systems must stay controllable, usable, and expressive in real contexts.

EXPERIENCE

Founder & Full-Stack Developer

2024-Present

Freelance Web Studio, Sheffield

- Own projects end to end for local businesses: client discovery, design, full-stack build, and deployment with React, Tailwind, GSAP, and Cloudflare/Vercel.
- Deliver responsive sites and SEO; developed B2B sales, proposal writing, and client-relationship skills.

Software Engineer

2026

Early-Stage AI Startup, London · Short-term

- Built an internal API to consolidate disjoint systems, and rewrote several Cloudflare Workers, including their GitHub syncing.
- Delivered a UI dashboard, a CLI, and an MCP server to unify their internal tooling.

Missions Engineer

2023-2024

Project Falcon, IMechE Student Team

- Built the Flight Ground Control System (FGCS) in Node.js and Python for a fixed-wing quadcopter: live telemetry monitoring, recording, and display.

EDUCATION

BSc Computer Science

Graduating 2026

The University of Sheffield

On track for first-class honours. Modules across AI, software engineering, and mathematics. Dissertation: MotionGen.

A-Levels and GCSEs

2016-2023

John F. Kennedy School and Sixth Form, Hemel Hempstead

A-Levels: Maths (A*), Computer Science (A), Further Maths (B), Physics (B).
10 GCSEs including Maths, English, and Computer Science.

HONOURS & AWARDS

HackSheffield 9

1st place, Best Repository · 2024

Snapchat Lens Competition

Lens used over 1M times, £1,500 prize · 2022

Dragon's Apprentice

Led the winning team, growing £100 into over £1,000 for charity, and won the creativity prize

Engineering You're Hired

3rd place, swarm-robotics pipe-inspection concept · 2025

SELSA Leadership Academy

Selected to lead three interdisciplinary engineering teams

Hackathon Category Win

2026

Regional Public Speaking

Runner-up

INTERESTS

- Bouldering to f7B+ across the Peak District, Lake District, Portland, and Fontainebleau
- Charity hitchhiking: Sheffield to Porto and Bratislava, and Croatia to Morocco
- Judo, competing nationally at BUCS
- Grade 8 guitar

SELECTED PROJECTS

MotionGen

Dissertation · 2026

Free, open-source Unity plugin that runs motion-generation models fully locally inside the Unity editor, with an accessible workflow and advanced editing features. Being prepared for release on the Unity Asset Store, pending University approval.

Unity · C# · Generative AI · Local inference

HabitFlow

Live · 2026

Full-stack habit and goal-tracking platform with social feeds, AI-powered insights via Google Gemini, and a GitHub-style consistency heatmap.

React · TypeScript · Express · PostgreSQL/Neon · Firebase Auth

Neighbourly

Live · 2026

Map-based app matching neighbours who need help with nearby volunteers, with real-time chat. Grew out of our HackSheffield 9 winning build (1st place, Best Repository).

SvelteKit · Flask · PostgreSQL · Google Maps · Auth0

AlexNet Transfer-Learning Classifier

Graded 95% · 2026

Fine-tuned AlexNet and selectively re-trained specific convolutional layers, then analysed training strategies, layer behaviour, and performance.

Python · PyTorch · CNNs · Transfer learning

3D Rasterizer Engine

2023

A full software 3D graphics pipeline built from scratch in Python, including projection, rasterization, z-buffering, and .obj model loading.

Python · Rasterization · Computer graphics

MNIST From-Scratch Classifier

2024

A complete ML pipeline built from scratch with custom SVM and KNN, PCA, and augmentation, reaching 99.8% accuracy on noisy test data.

Python · NumPy · SVM · KNN · PCA

Offline AI Mobile App

In progress

On-device LLM inference using llama.cpp and quantised GGUF models, with offline maps and survival guides for use without connectivity.

Python · llama.cpp · GGUF · On-device inference

Modelling Natural Systems and Reinforcement Learning

Graded 95% and 100% · 2026

Modelled and analysed dynamical systems, including the Duffing oscillator and a predator-prey agent-based model, and explored tabular Q-learning.

Python · NumPy · Agent-based modelling · Reinforcement learning